

Jagraj Gill

Canadian Citizen | jsg51@sfu.ca | github.com/jagrajg | jagrajgill.tech | linkedin.com/in/jagrajgill

Education

Simon Fraser University – Bachelor of Science in Computing Science

Dec 2027

Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, HTML, CSS, x86 Assembly

Frameworks: React, Node.js, Express.js, REST APIs, GraphQL, FastAPI, CI/CD, Agile

Databases & Tools: Git, Docker, MongoDB, MySQL, SQLite, AWS S3, Linux, Confluence, TCP/IP

Experience

Software Engineer Intern, Savi Finance – Toronto, ON

May 2025 – Dec 2025

- Developed the core AI logic and MongoDB schema that interpreted prioritized tasks and routed them to third-party fulfillment APIs
- Built feature parity across React and React Native codebases by extending a shared GraphQL API with mobile-specific schema for 1000+ users
- Reduced image load failures by 10% with an AWS S3 and GraphQL upload feature for reliable previews
- Improved release safety with GitHub Actions checks flagging package/configuration changes before deployment

Projects

HTTP Web Server  – C++, TCP/IP, HTTP

- Built a low-level HTTP server in C++ using POSIX sockets to handle TCP connections, parse HTTP requests, and serve static HTML, CSS, JavaScript, and image files
- Implemented GET, POST, and DELETE methods with status codes and headers for spec-compliant responses
- Built MIME detection, POST parsing, logging, and error handling for reliable, debuggable request processing
- Added poll()-based non-blocking I/O to support concurrent browser and curl clients without threads

AI-Powered Study Assistant  – Python, JavaScript, FastAPI, Gemini API

- Built a pipeline using PyPDF and a structured LLM prompt to convert PDF text into flashcards, automating study material generation from unstructured documents
- Designed a strict JSON schema requiring the AI to return all flashcard fields, reducing incomplete output
- Built Python validation to catch empty fields and malformed content before they reached the flashcards
- Developed an interactive JavaScript flashcard interface with flip animations and navigation controls

Smart Environmental Monitoring System  – C, JavaScript, SQLite, FastAPI

- Built a full-stack IoT monitoring system that streamed live temperature and humidity readings from an embedded sensor device to a web dashboard
- Developed a FastAPI backend that ingested JSON sensor data over Wi-Fi, storing readings in SQLite and powering a live JavaScript dashboard with 24-hour trend views
- Designed fault-tolerant data ingestion that handled connection failures without crashing or failing silently

Portfolio Performance Tracker  – Java, Swing

- Built a Java Swing application where users could log stock holdings to visualize portfolio growth trends
- Calculated gain/loss for each stock holding, rendering results as graphs to compare performance across assets
- Added input validation to reject non-numeric or unrealistic entries, preventing corrupted portfolio data

Wireless Robotic Arm  – C, ESP32, Embedded Systems

- Built a real-time 4-DOF robotic arm control system using dual ESP32 devices over a peer-to-peer wireless link
- Designed a data pipeline that mapped raw analog sensor readings to per-joint servo angles, applying exponential smoothing and range constraints to prevent erratic or unsafe motion